

Finish the project, good luck

Future Tasks:

- Landing page panel (maybe (definitely))
- GIS Data
 - Area selection: The Landing Page should display different area boundaries. These boundaries would be labeled on the Landing Page Panel and would contain a certain amount of trailpins. An area would highlight when clicked on, and would zoom in and display all trailpins within its boundaries.
 - Boundaries: Website administrators should be able to add, edit, and remove boundaries around specified areas. Additionally, the entire park should have a hard boundary that users should not be able to scroll past.
- Multi-select: Multiple different trailpins should be selectable at once. Information about these trails would be displayed on the Landing Page Panel and moving over to the dashboard would display relevant data for all selected pins.
- Daily Report Creation: Trail statistics should be interpreted in the form of a daily report.
 - Advanced Reports: Reports should answer key questions (e.g. “Busiest trail yesterday,” “Peak times for Trail X”).
 - These reports could potentially be viewable somewhere on the site, or sent out to a list of users every morning.
 - Implementation Wise: This would require creating a lambda that runs every morning that takes in the data needed for the report and generating it. It would store the report somewhere with a way for the application to retrieve or send it to the list of users. (Potential Technologies: Lambda, CloudWatch, DynamoDB, S3, APIGateway (application retrieval), Cognito (user sending), SES (user sending))
- Permission/Role changes: Renaming the roles and giving each of them different permissions that match the website’s current functionality.
 - An approach that might be more exciting to pursue is a set of permissions that can be granted on a case by case basis, or role by role basis. Administrators with the permissions to do so could be able to create new roles with a specific set of permissions, and apply them to users. This way permissions can be better controlled by a smaller group of top level administrators, and specific permissions for workflows like device provisioning can be granted without access to other sensitive functions like user configuration.
 - A further enhancement for this feature would be creating a system for managers to request permissions that they don’t currently have, with it sending a request and reason for the request that someone with more permissions could approve.
 - Implementation Wise: This would likely mean moving from the existing cognito user-group based authentication to a system that stores and manages roles itself, likely with dynamodb, with each permission being predefined and added/removed from a given role. A user’s roles would then become attributes directly attached to the cognito user. (Potential Technologies: Cognito, APIGateway, DynamoDB, Lambda)

- Accessibility controls: Color blind support, as well as text-to-speech functionality across all pages, night mode
- Alternate dashboard data views
 - Potential alternate views for the graph in the dashboard, and enhancements to the current view
 - A button to aggregate all currently selected trails into a singular line
 - Upper and lower percentile lines for the default graph view based on the historical averages and selected granularity
 - The ability to overlay multiple years of data on the same graph to directly compare different years
 - A view that changes the data to be
 - A “heatmap” of visitors counts by day and week to see more concentrated peaks. Like the graphs with the colored squares where each day is a square and is colored differently depending on the count.
- User Preferences: Many users like being able to set preferences or settings.
 - Dark/Light Mode: Some way to change the page from dark to light and vice versa
 - Change password or account information
 - Delete account
 - Linking accounts (google, github, etc)
 - Display Name
 - Bio
 - Multifactor authentication
- Bug Reporting Mechanism: Things can be missed, and normal users don't have access to the DevOps team. There should be some way to report a bug.
 - Should be easy to find
 - Should be rate limited or otherwise validated
 - Should contain sections for:
 - Where the bug was
 - The steps for reproduction
 - A section for if the user wants to be notified on bug fixing progress
 - Description of bug